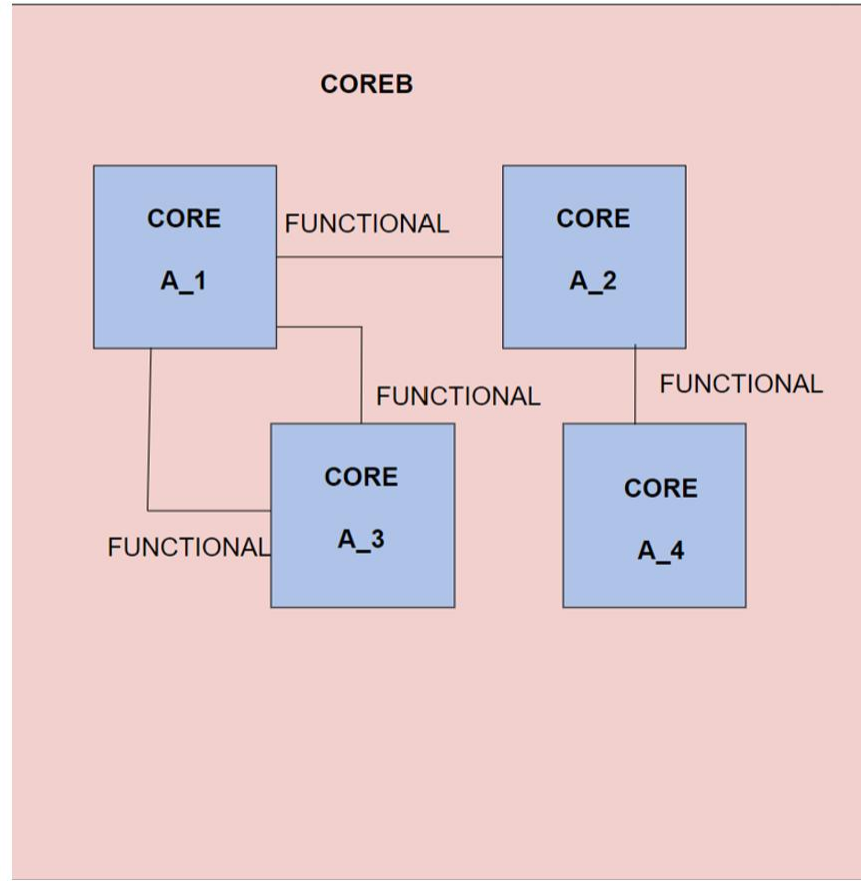


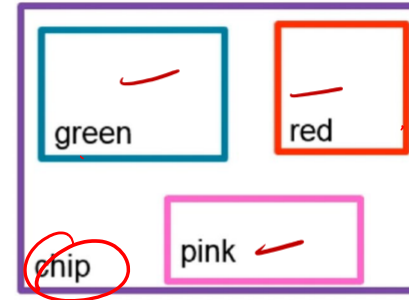
Project in Short

DESIGN



Hierarchical

DFT insertion is performed bottom up for each of the green, red and pink hierarchical layout regions, and then DFT insertion is performed at the chip level.



→ DFT for core A ✓✓

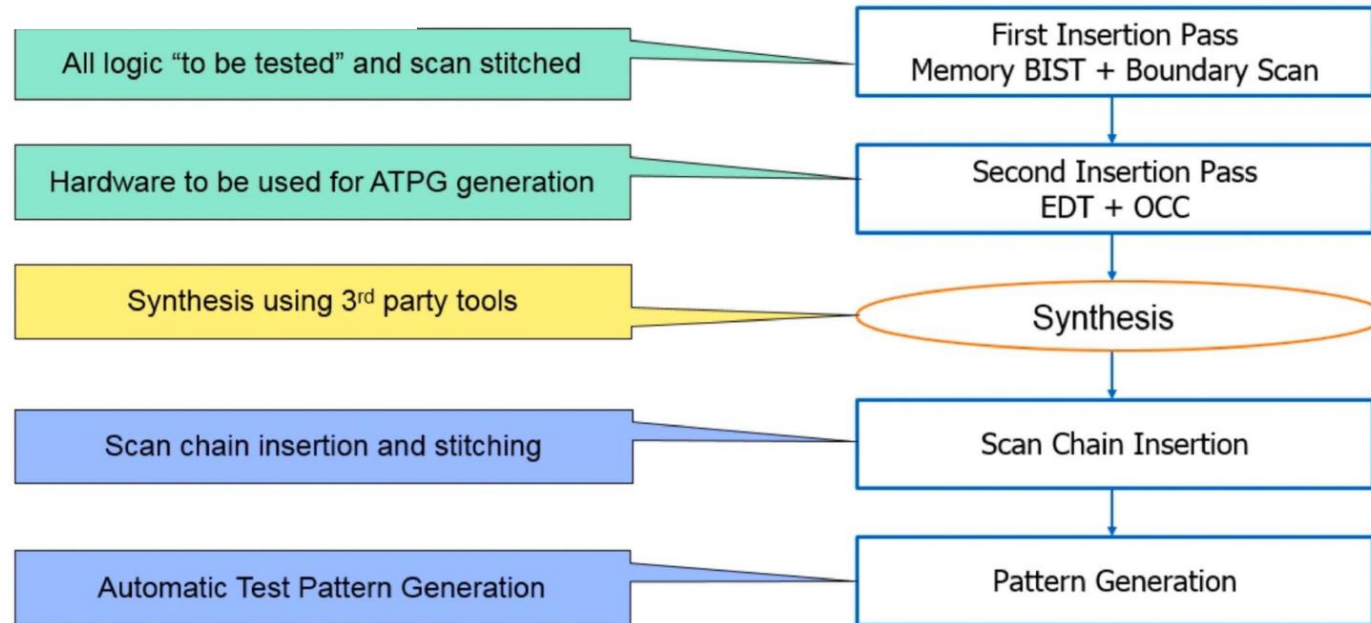


DFT for core B.



Pre-Scan DFT Insertion

DFT insertion is performed on pre-scan inserted netlist.



Pre Scan Insertion Flow for RTL

Lab 3 intest ATPG

↳ core A intest

↳ To test the internal logic of core A.

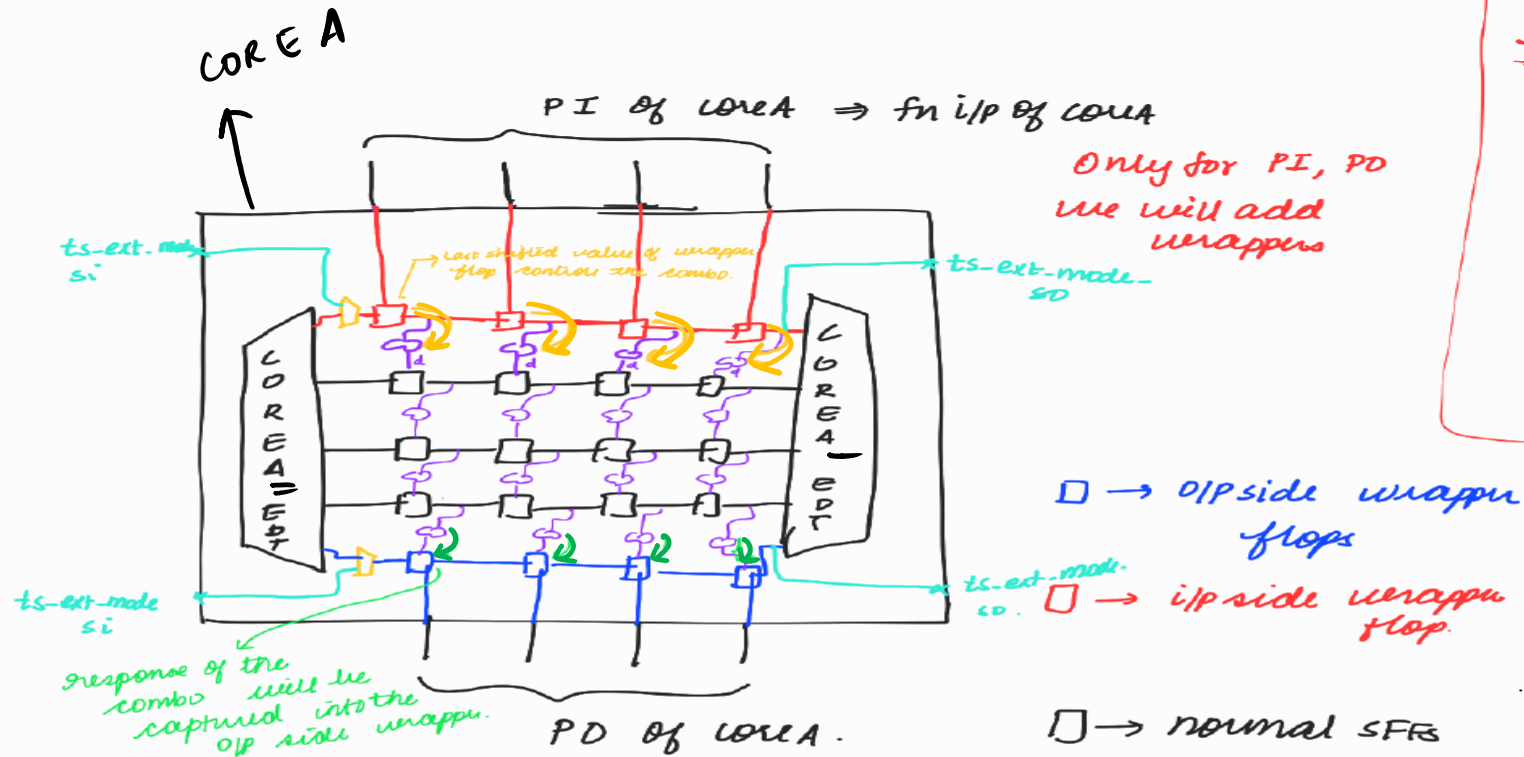


This is tested in the intest run of core A
Lab 3 intest.

CORE A's EDT —→ control CORE A's normal SFFs
 → CORE A's unappn flops

↳ i.e. Lab 3 intest:-
patterns are generated
in order to test the internal
logic of core A.

Lab 3 Intert Core A Intert.



Role of ^{Core A's} wrappers during Intert of Core A :-

i/p side wrapper chains launch data into inside logic
(dep. in yellow)

o/p side wrapper wrapper chains capture data from inside logic
(dep in green)

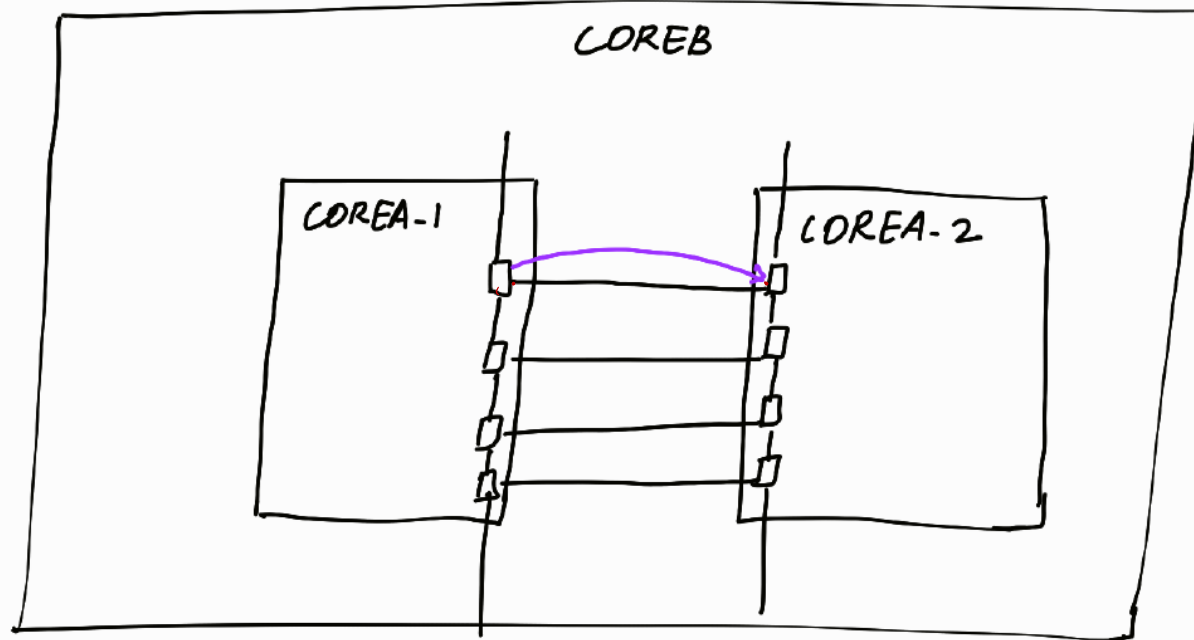
During this time, the i/p sequence will be shifted into the wrapper chains by Core A EDT.

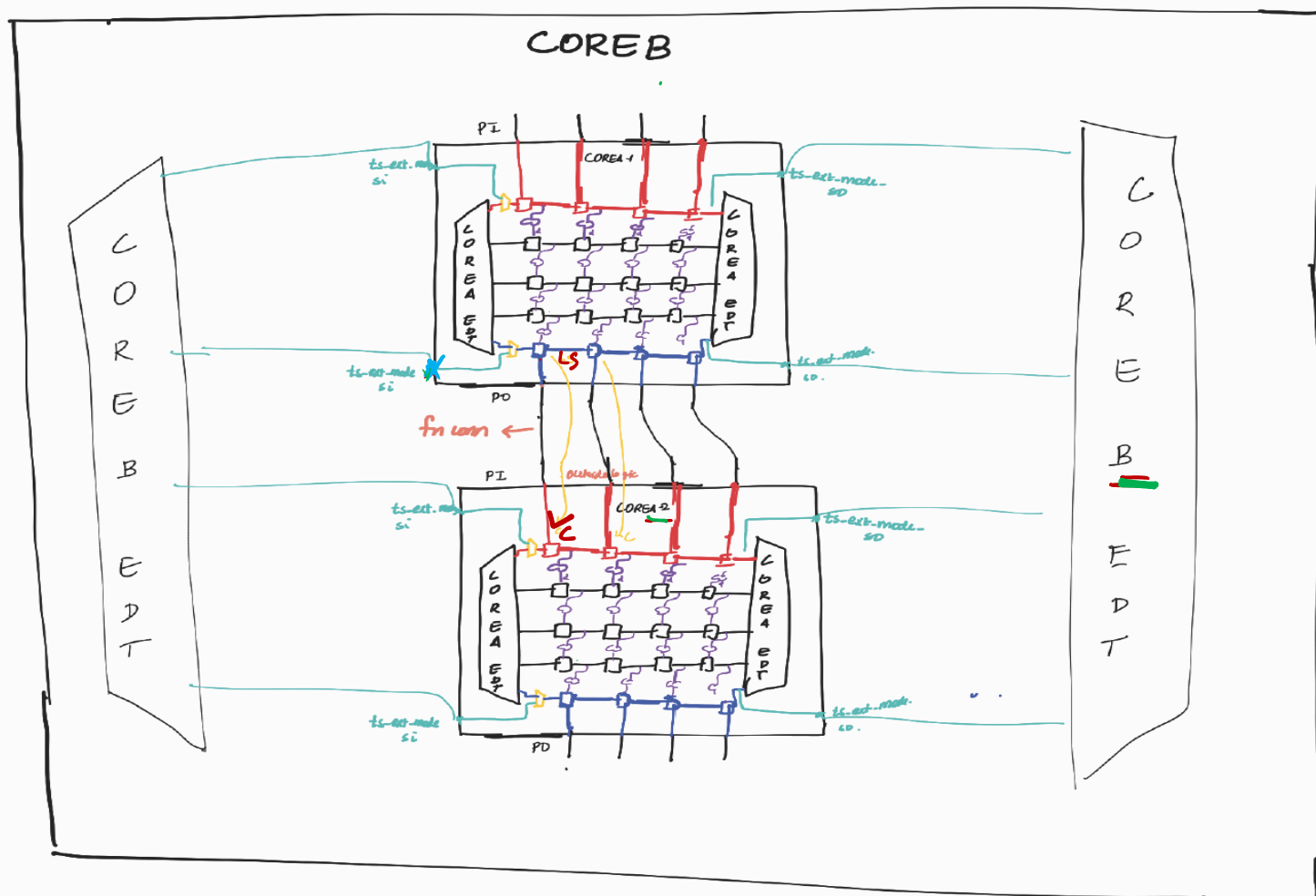
Normal SFFs also, they will be doing their usual operation (shift, cap, shift out)

Note:-

The actual testing of interconnections will happen during coreB level.

actual extent of coreA.





* Last shifted value of core-1's O/P side wrapper (blue column) will be captured by core-2's i/p side wrapper (red column).

If there was some defect in the interconnection, then wrong value will be captured.

COREB level

Actual testing of interconnection b/w coreAs will happen. see *

↳ Actual extent of coreA.

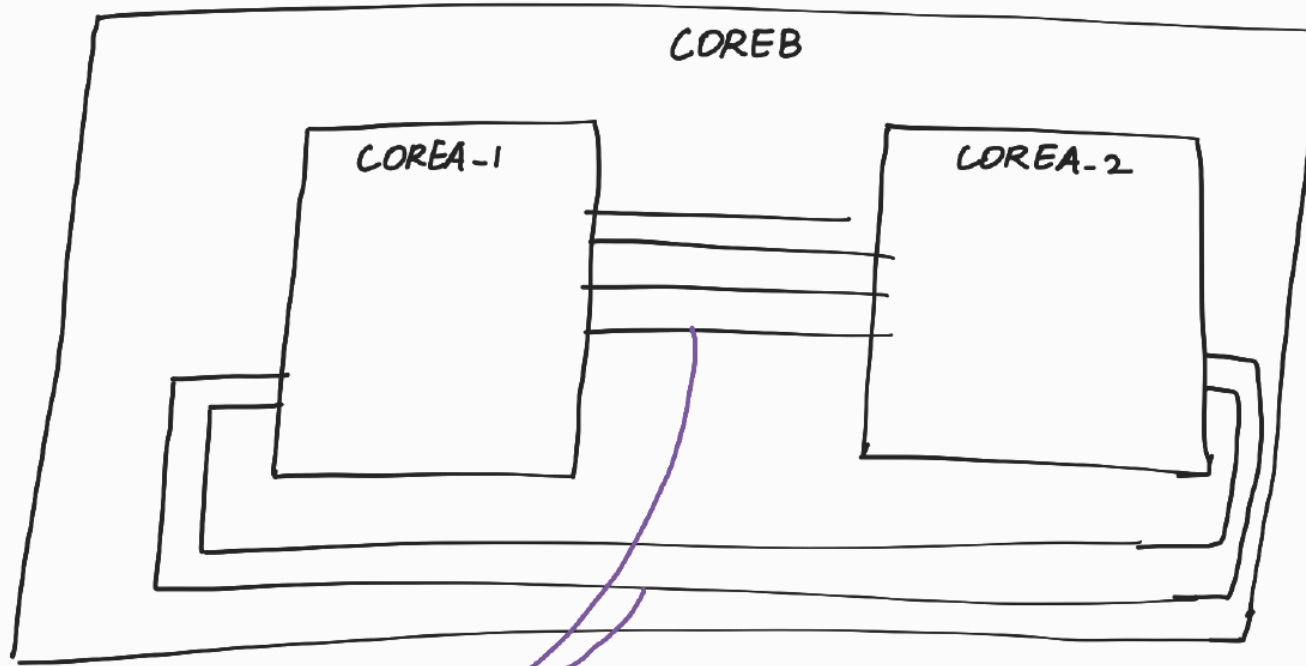
During this time (COREB level)

↳ COREAs will be in extent

↳ COREB's EDT will only be shifting the values into

COREA's wrapper flops.

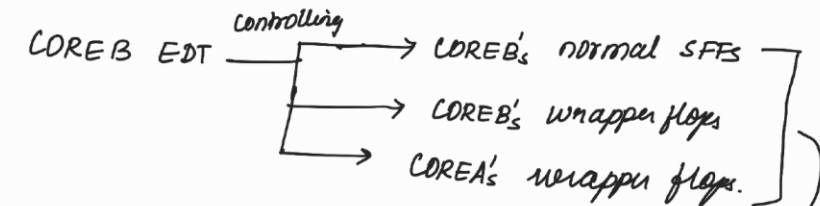
ATPG interest
Interest of CORE B



interconnections + core logic of COREB
will be tested $\Rightarrow$ interest of COREB.

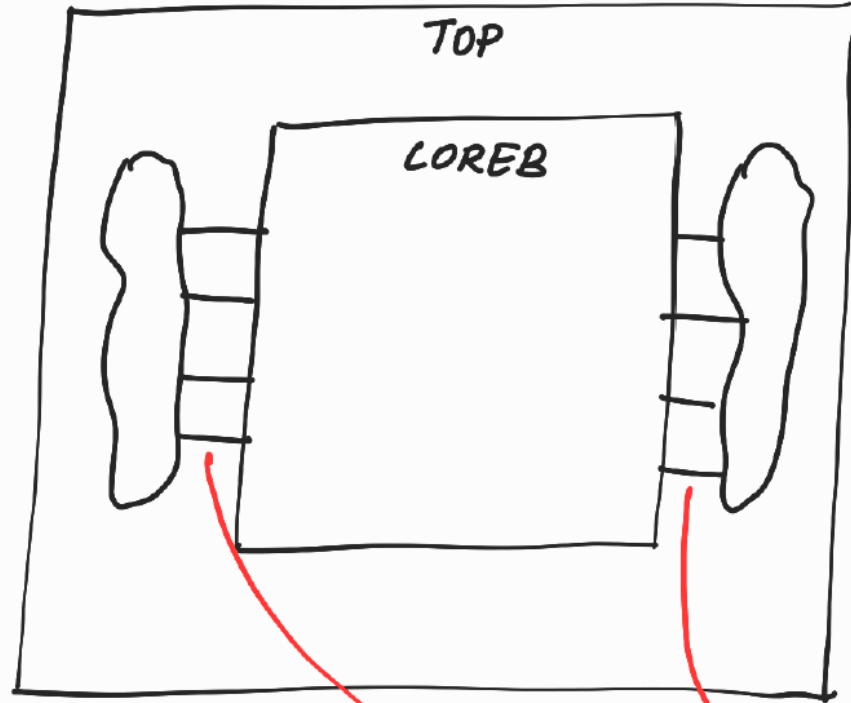
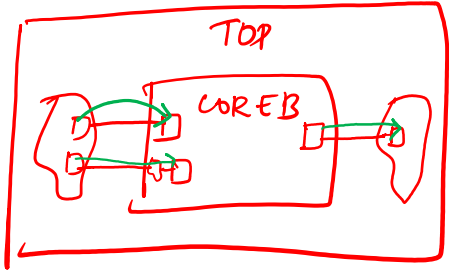
Logic inside coreB
+
Interconnections b/w coreA_S

will be tested
i.e; patterns
will be
generated to
test them



← COREB's EDT will only be shifting value into them.

Actual ext test of core B. $\Rightarrow$ happens at the top level.



These interconnections
will be tested during TOP
level.

PATTERN RE-TARGETING

Re-applying already generated coreA intest patterns from CoreB level instead of generating it again.

After top insertion, coreA intest patterns will be again re-applied from top level also.

Why pattern retargeting?

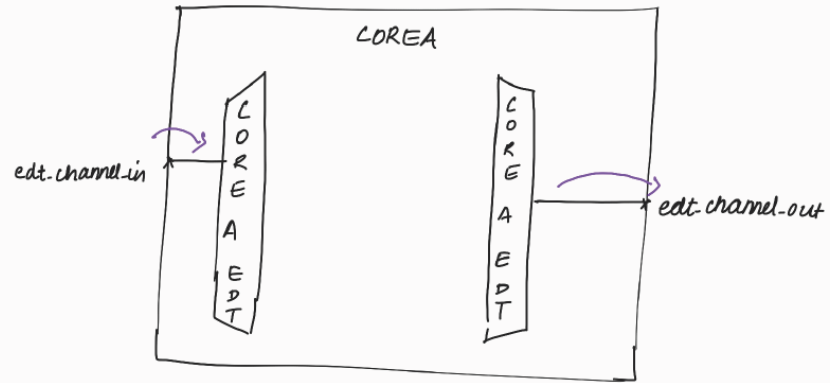
Finally at the tester level, everything has to be applied from top level.

Why only coreA intest?

CoreA extest (interconnections) are already tested at the coreB level and patterns are generated from coreB level.

COREA level :-

COREA intent :- To test the logic inside COREA



COREB level :-

Re-applying
already
generated
coreA intent
patterns
from
coreB level
↓
Pattern
re-targetting.

